

Research Paper Writing 1

Session 01: Introduction

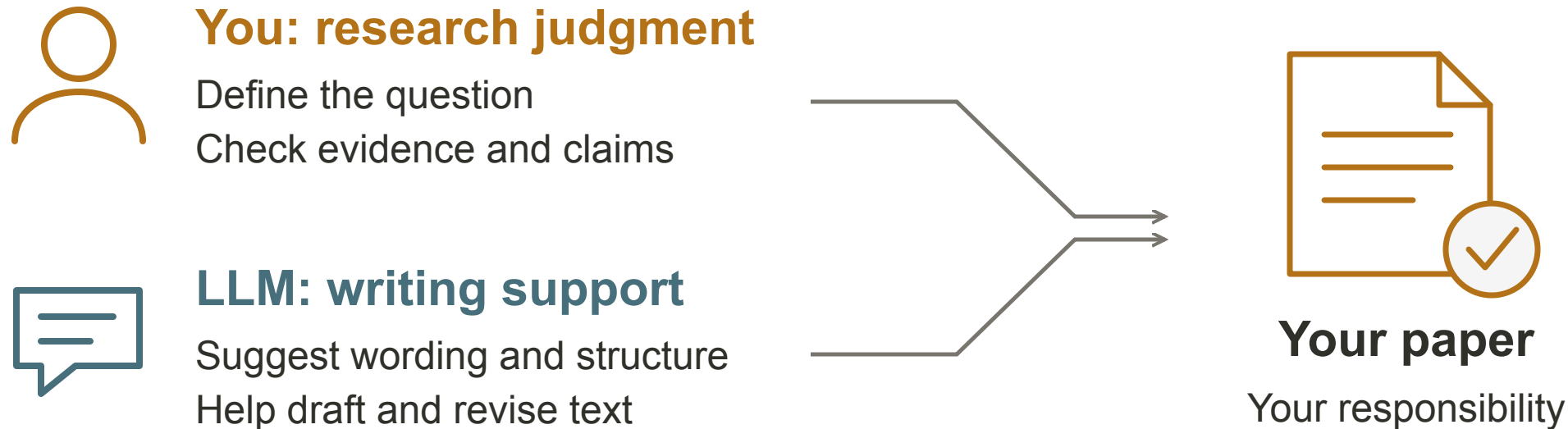
Academic Year 2026, Q3 and Q4

RPW1 Team

University of Aizu

Why RPW1 in the era of LLM?

An LLM can help with text. You still need to judge the research.



RPW1 teaches you to build and check a clear research argument.

Today's topics

- Course overview and student survey
- Research papers and why we write them
- The research and writing process
- Peer review and revision
- Clear writing and course assignments

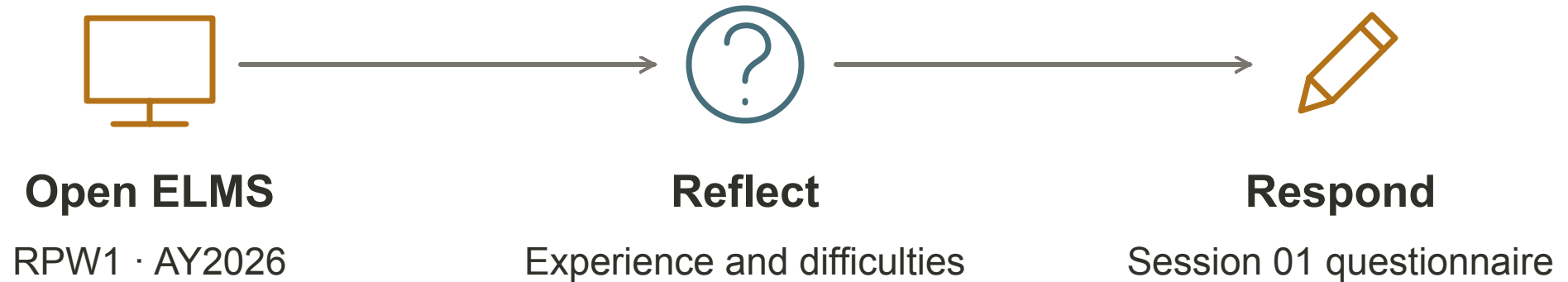
Teaching team

The teaching team included:

- KACHI Yasuyuki: Mathematics
- ROY Debopriyo: Linguistics
- VIGLIETTA Giovanni: Theory of Informatics
- NASSANI Alaeddin: Augmented Reality
- DANG Nam Khanh: Computer Architecture

We also have several invited lecturers!

Session-01 survey



About 10 minutes

[→ Open ELMS](#)

RPW1 Survey



Research experience

89% had experience
11% had no experience



Paper writing

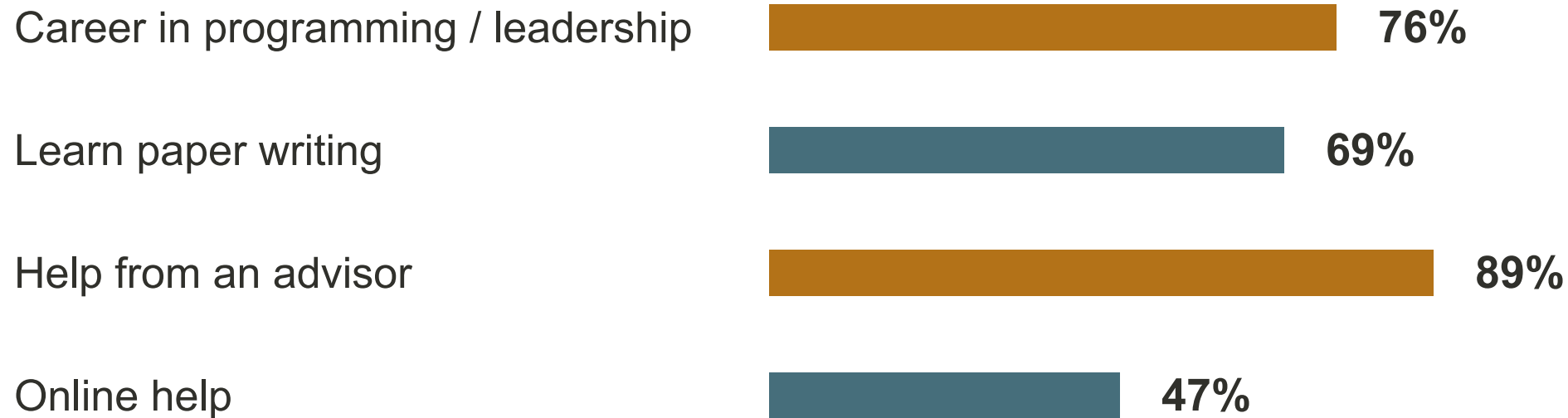
62% writing or had written
38% had never written

2025 survey: 45 responses · Writing difficulty: 7.22 / 10

These results describe the 2025 class.

RPW1 Survey: goals and support

2025 survey · 45 responses



Multiple answers were allowed

These results describe the 2025 class.

Common writing difficulties

- Explaining an idea clearly
- Organizing sections and connecting ideas
- Academic English and word choice
- Choosing a research topic
- Using LaTeX

Course goals



- Your supervisor guides your research.
- This course focuses on research writing, rather than general English lessons.

Research fields

The course uses examples from computer science and engineering.

- Computer systems and networks
- Security, software, and databases
- Theory and algorithms
- AI, graphics, and multimedia
- Human–computer interaction

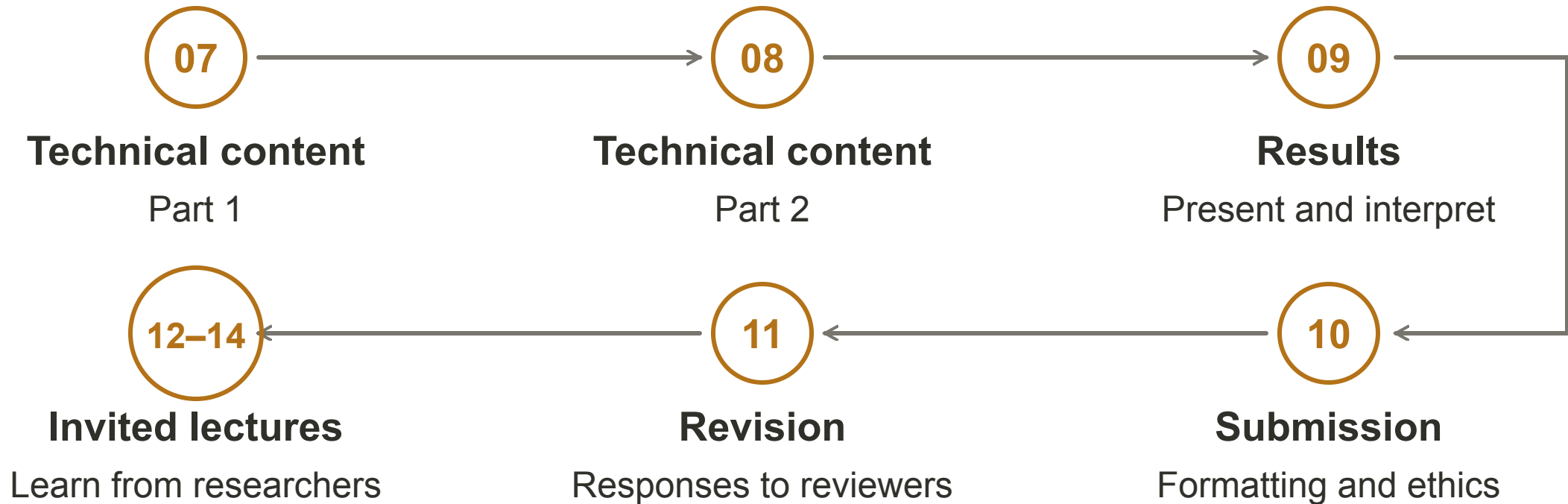
Apply the writing principles to your own field.

Course plan: Sessions 01–06



Check ELMS for the confirmed order and dates. Topics and dates may change.

Course plan: Sessions 07–14



Check ELMS for the confirmed order and dates. Topics and dates may change.

Course information in ELMS

Use the **AY2026 RPW1 course page** for:

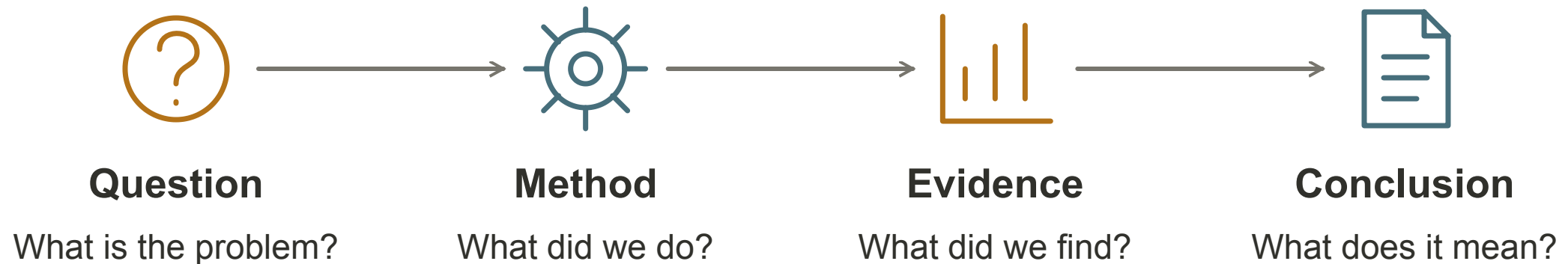
- Class dates, rooms, and schedule changes
- Attendance and grading rules
- Plagiarism and generative AI rules
- Slides, questionnaires, and assignments

The AY2026 plan used periods 9–10, **17:00–18:40**, Mon, Tue, or Wed.

Check the current schedule before each class. We will also notify via ELMS, please check your email.

What is a research paper?

A research paper explains a question and what the authors learned.
It tells readers:



Example: A systems paper

Pollux: Co-adaptive Cluster Scheduling for Goodput-Optimized Deep Learning

The first page includes a title, authors, abstract, and introduction.

How does the abstract explain the problem and result?

Pollux: Co-adaptive Cluster Scheduling for Goodput-Optimized Deep Learning

Aurick Qiao^{1,2} Sang Keun Choe² Suhas Jayaram Subramanya² Willie Neiswanger^{1,2}
Qirong Ho¹ Hao Zhang^{1,3} Gregory R. Ganger² Eric P. King^{4,1,2}

¹Petuum, Inc. ²Carnegie Mellon University ³UC Berkeley ⁴MBZUAI

Abstract

Pollux improves scheduling performance in deep learning (DL) clusters by adaptively co-optimizing inter-dependent factors both at the per-job level and at the cluster-wide level. Most existing schedulers expect users to specify the number of resources for each job, often leading to inefficient resource use. Some recent schedulers choose job resources for users, but do so without awareness of how DL training can be re-optimized to better utilize the provided resources.

Pollux simultaneously considers both aspects. By monitoring the status of each job during training, Pollux models how their *goodput* (a metric we introduce to combine system throughput with statistical efficiency) would change by adding or removing resources. Pollux dynamically (re-)assigns resources to improve cluster-wide goodput, while respecting fairness and continually optimizing each DL job to better utilize these resources.

In experiments with real DL jobs and with trace-driven simulations, Pollux reduces average job completion times by 37–50% relative to state-of-the-art DL schedulers, even when they are provided with ideal resource and training configurations for every job. Pollux promotes fairness among DL jobs competing for resources, based on a more meaningful measure of *weight* (job progress), and reveals a new opportunity for reducing DL cost in cloud environments. Pollux is implemented and publicly available as part of an open-source project at <https://github.com/petuum/adaptDL>.

1 Introduction

Deep learning (DL) training has rapidly become a dominant workload in many shared resource environments such as datacenters and the cloud. DL jobs are resource-intensive and long-running, often demanding distributed execution using expensive hardware devices (e.g. GPUs or TPUs) in order to complete within reasonable amounts of time. To meet this resource demand, dedicated clusters are often provisioned for deep learning [31, 67], with a scheduler that mediates resource sharing between many competing DL jobs.

Existing schedulers require users to manually configure their jobs, which if done improperly, can greatly degrade training performance and resource efficiency. For example, allocating too many GPUs may result in long queuing times and inefficient resource usage, while allocating too few GPUs may result in long runtimes and unused resources. Such decisions are especially difficult to make in a shared-cluster setting, since optimal choices are dynamic and depend on the cluster load while a job is running.

Even though recent elastic schedulers can automatically select an appropriate amount of resources for each job, they do so blindly to inter-dependent training-related configurations that are just as important. For example, the *batch size* and *learning rate* of a DL job influence the amount of computation needed to train its model. These optimal choices vary between different DL tasks and model architectures, and they have strong dependence on the job's allocation of resources.

The amount of resources, batch size, and learning rate are difficult to configure appropriately without expert knowledge about both the cluster hardware performance and DL model architecture. Due to the inter-dependence between their optimal values, they should be configured jointly with each other. Due to the dynamic nature of shared clusters, their optimal values may change over time. This creates a complex web of considerations a user must make in order to configure their job for efficient execution and resource utilization.

How can a cluster scheduler help to automatically configure user-submitted DL jobs? Fundamentally, a properly-configured DL job strikes a balance between two often opposing desires: (1) *system throughput*, the number of training examples processed per wall-clock time, and (2) *statistical efficiency*, the amount of progress made per training example processed.

System throughput can be increased by increasing the batch size, as illustrated in Fig. 1a. A larger batch size enables higher utilization of more compute resources (e.g., more GPUs). But, even with an optimally-tuned learning rate, increasing the batch size often results in a decreased statistical efficiency [46, 57]. For every distinct allocation of GPUs, there is potentially a different batch size that best balances increasing

Example: A machine learning paper

Attention Is All You Need

Look at the title, author list, abstract, and section headings.

How do they help you find the main idea?

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state-of-the-art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 21, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 11].

^{*}Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†]Work performed while at Google Brain.

[‡]Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Why research writing matters



Writing makes your reasoning visible.

Example: LLM training costs

Manager: "Training our large language model costs too much."

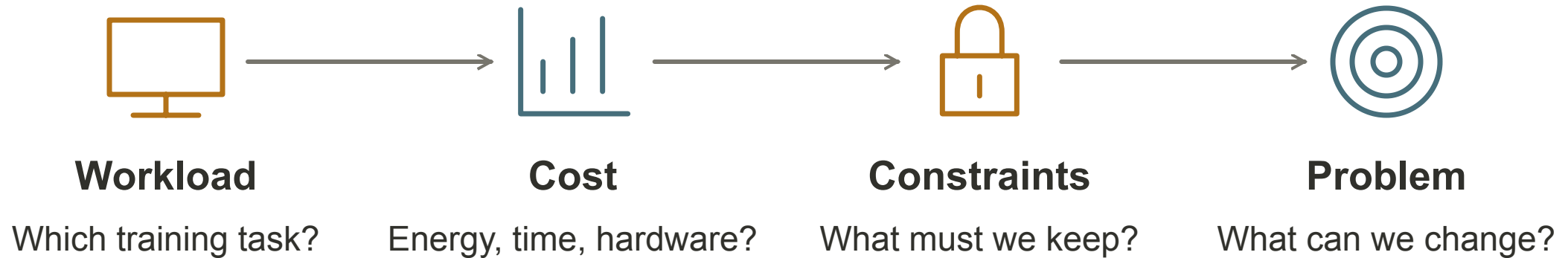
Possible replies:

- "Buy cheaper GPUs."
- "Rent computing resources."
- "Measure the costs first."

Which reply helps us understand the problem?



A clear problem



Measure before choosing a solution.

Evidence from earlier work



Read the study

Hardware and workload

Measurement method

Reported limits



Check the claim

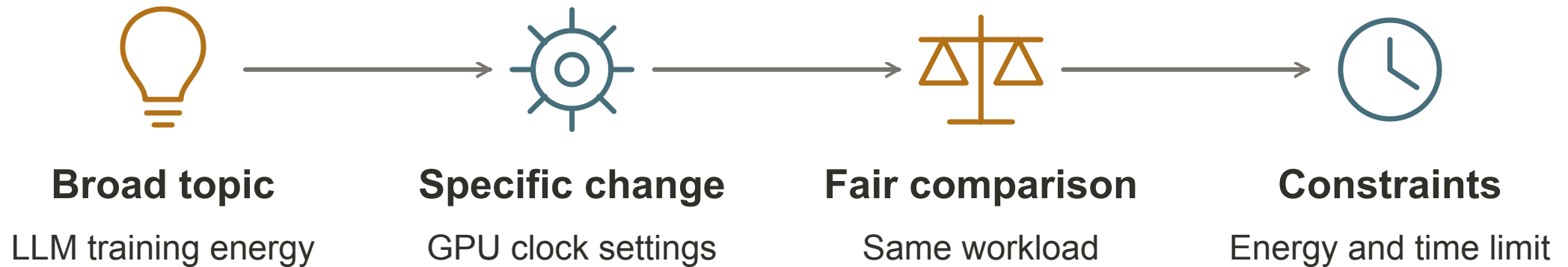
Training or inference?

Does it fit our question?

Can we trace the source?

A trusted source still needs careful reading.

A focused research question



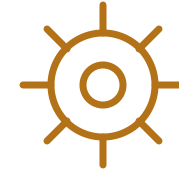
Can clock settings reduce energy while meeting a time limit?

A method and a fair test



Baseline

Same workload
Same hardware



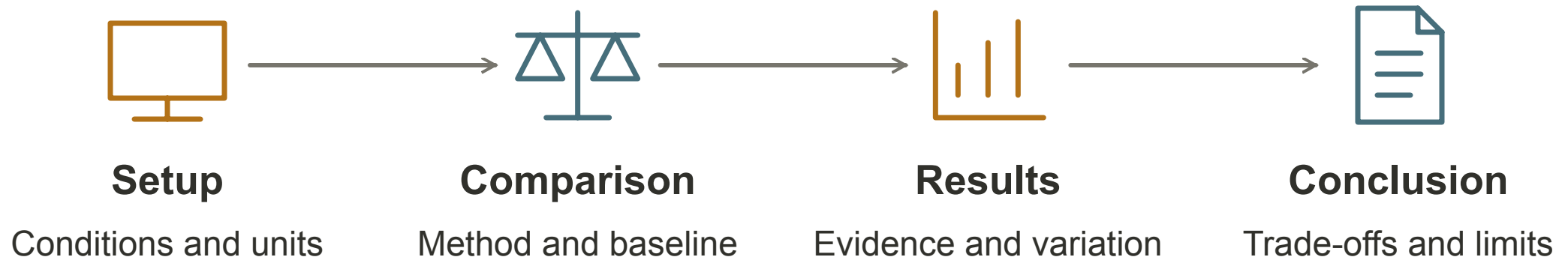
Proposed method

Same workload
Same hardware

Compare energy, training time, and model quality. Repeat tests.

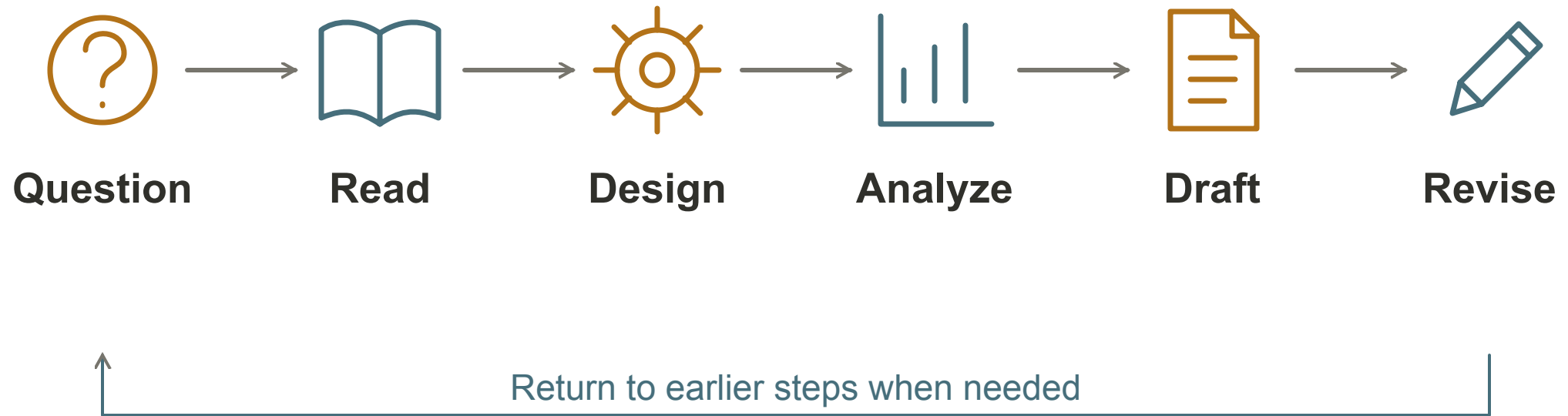
Explain where the method helps or fails.

Reporting the result



Keep the conclusion within the evidence.

The research and writing process



A useful research question

A useful question is:

- Clear enough for readers to understand
- Narrow enough for the planned study
- Possible to answer with available resources
- Important to the research community

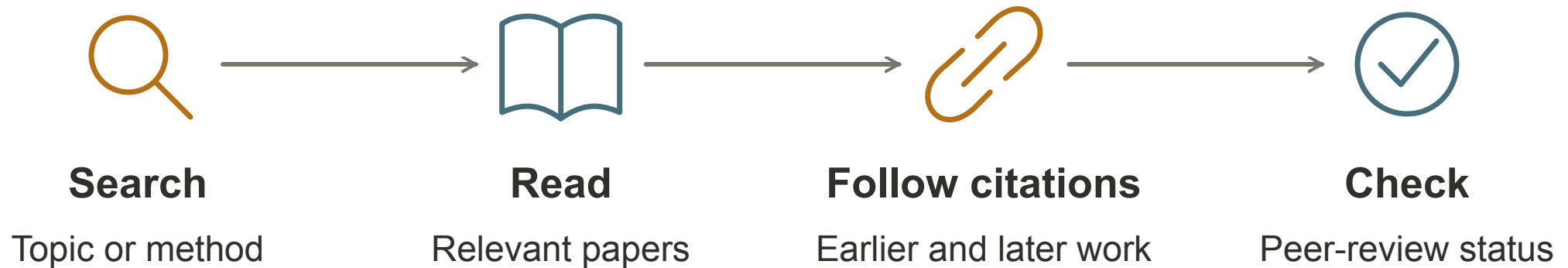
Activity: Write one question about your own research.
What evidence would answer it?

Literature review



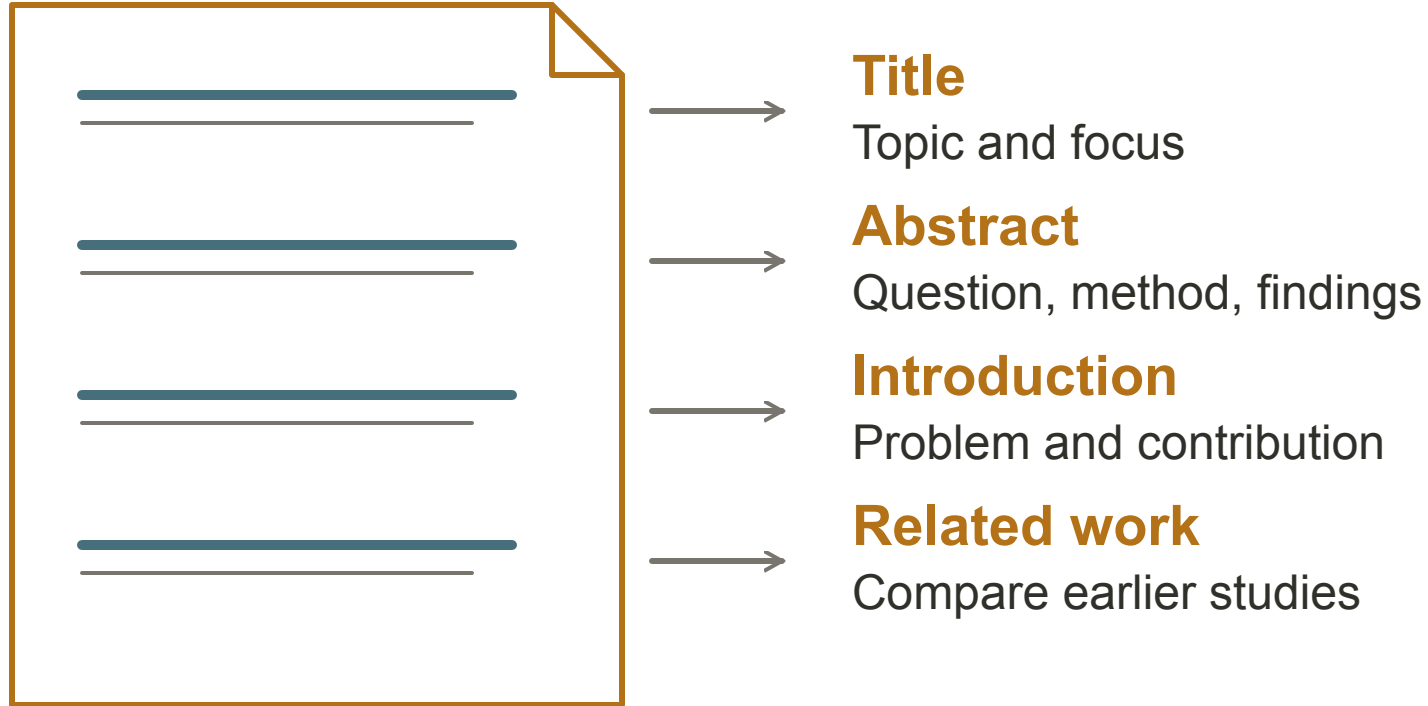
Explain how earlier work relates to your question.

Finding relevant papers



[Google Scholar](#) · [IEEE Xplore](#) · [ACM DL](#) · [arXiv](#)

Paper structure: The opening



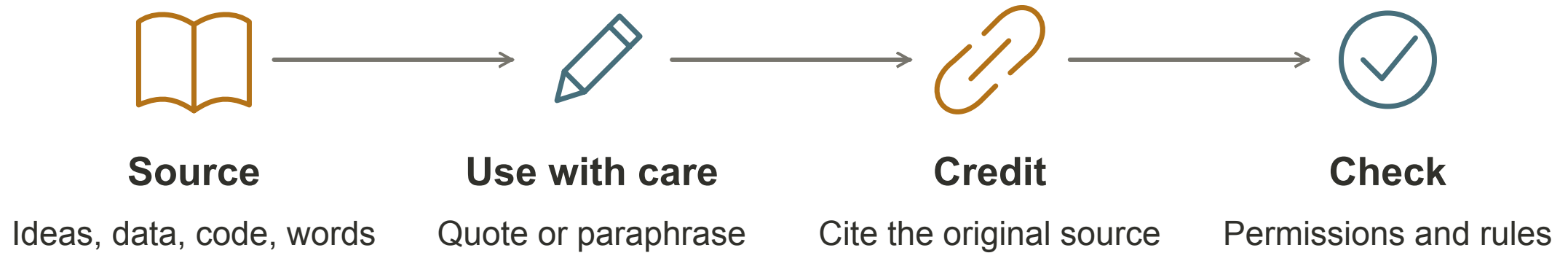
Section names and order vary by field and venue.

Paper structure: The main content



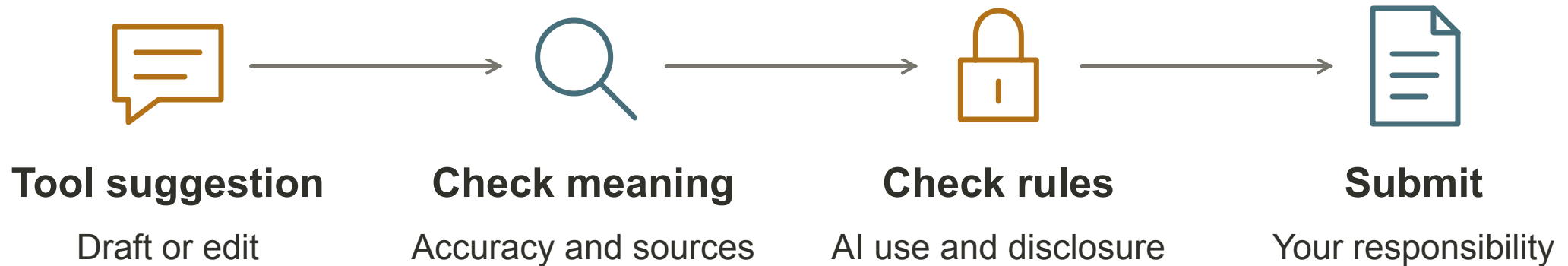
Some papers combine sections or use different names.

Citations and plagiarism



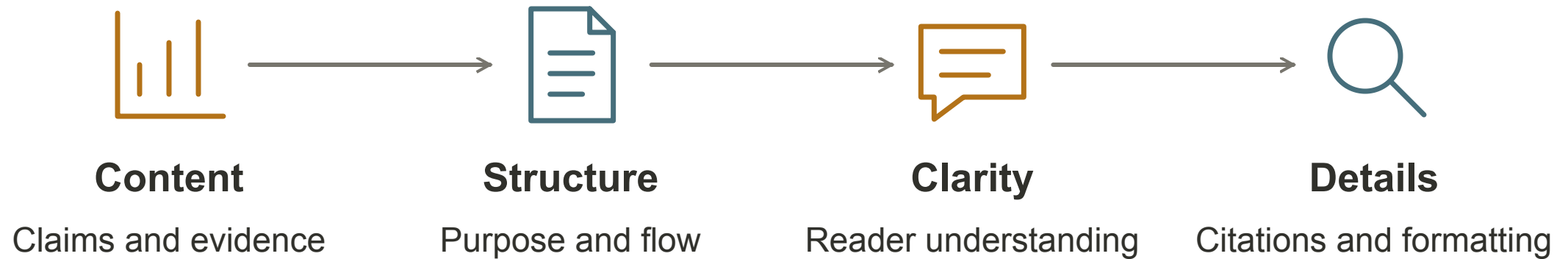
- Cite your own previous work when you reuse or extend it.
- Be cautious of AI hallucinations, and verify AI-generated claims and references.
- Permission may be needed for reproducing.

Writing tools and generative AI



You remain responsible for the work you submit.

Revision



Ask a reader to explain your main point back to you.

Common problems in a draft

- A broad problem with no clear research question
- A contribution that readers cannot identify
- Claims without enough evidence
- A list of papers without comparison or analysis
- Figures or explanations that are hard to follow
- Missing citations or incorrect formatting

Fix the main argument before polishing individual sentences.

Writing and reference tools

Task	Tools
Writing and formatting	Word, LaTeX, Overleaf
Managing references	Zotero, JabRef, Mendeley
Writing guidance	LLM (?)

Choose tools that fit your supervisor's workflow and the venue.
Check the output yourself.

Peer review

Peer review asks other researchers to assess a paper.

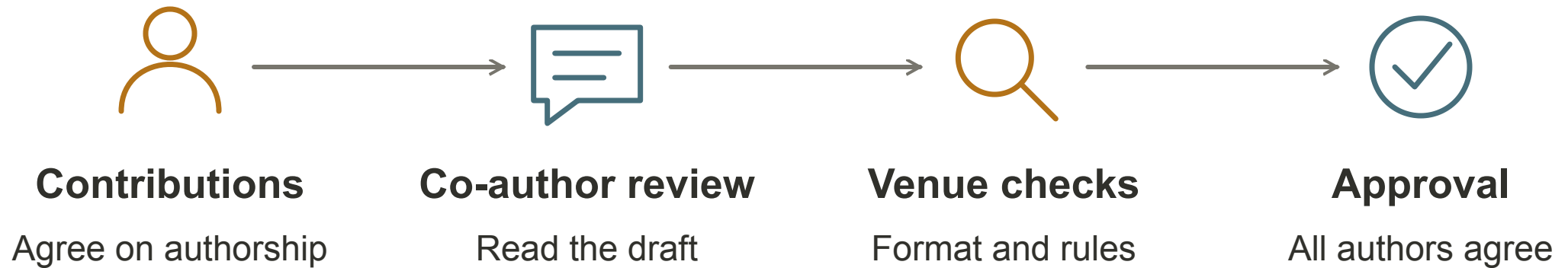
Reviewers examine:

- The question and contribution
- The technical approach
- The evidence and conclusions
- The clarity of the explanation

The editor or program committee makes the decision.

Source: <https://journals.ieeeauthorcenter.ieee.org/submit-your-article-for-peer-review/about-the-peer-review-process/> (checked 2026-10-01).

Before submission



Submission and review



The number of reviewers and steps depend on the venue.

Anonymous review and conflicts



Single-anonymous

Reviewer identities hidden
Author identities visible



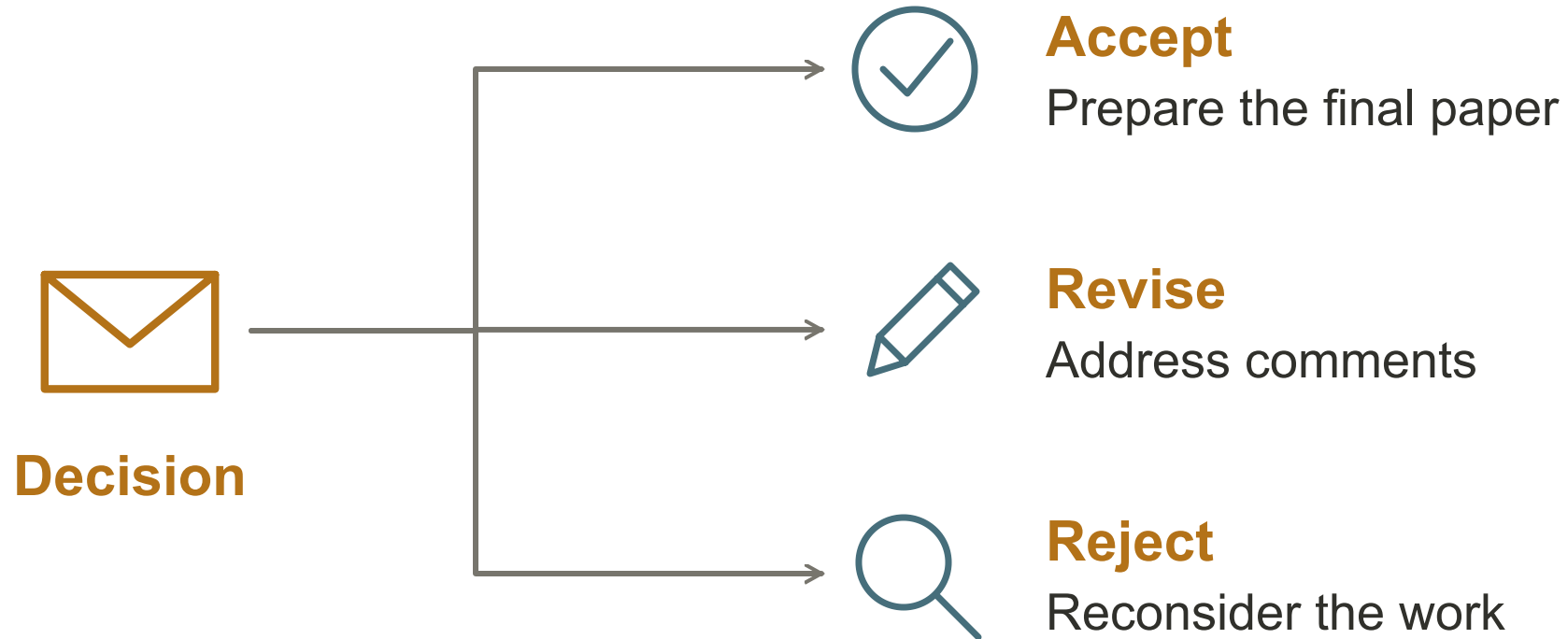
Double-anonymous

Reviewer identities hidden
Author identities hidden

Reviewers need relevant expertise and must disclose conflicts.

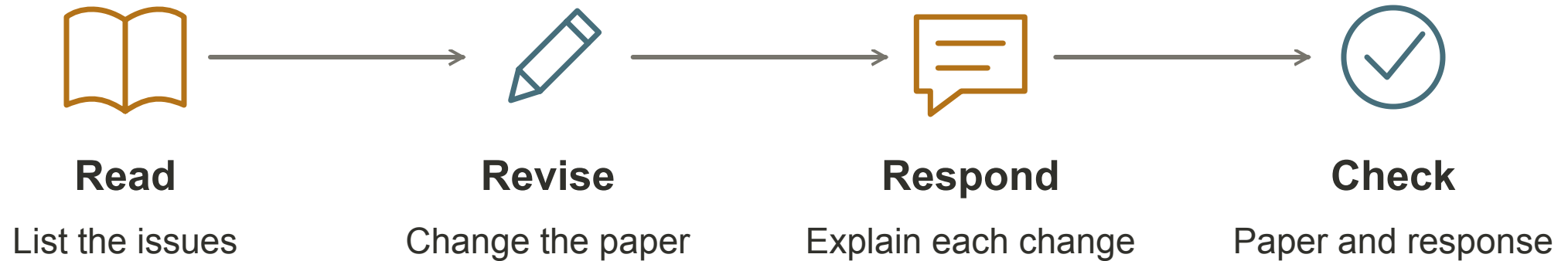
Follow the venue's specific rules.

Possible decisions



A revision request does not guarantee acceptance. Check the venue's procedure.

Responding to reviews



Support disagreements with reasons and evidence.

A useful review



Vague feedback

“The experiments
are weak.”

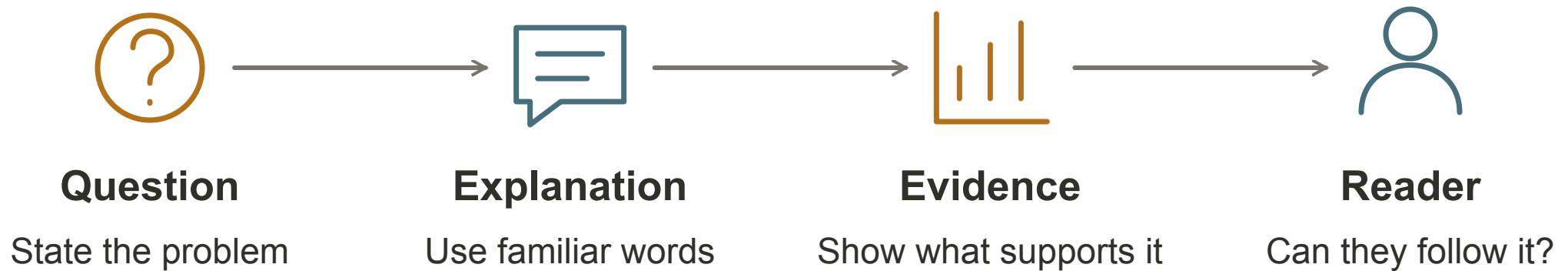


Useful feedback

“Test a second workload
to show whether the result
applies more widely.”

Be specific, fair, and constructive. Keep reviews confidential.

Clear scientific writing



Precision matters more than impressive language.

A precise sentence



Vague claim

“Our method
is much better.”



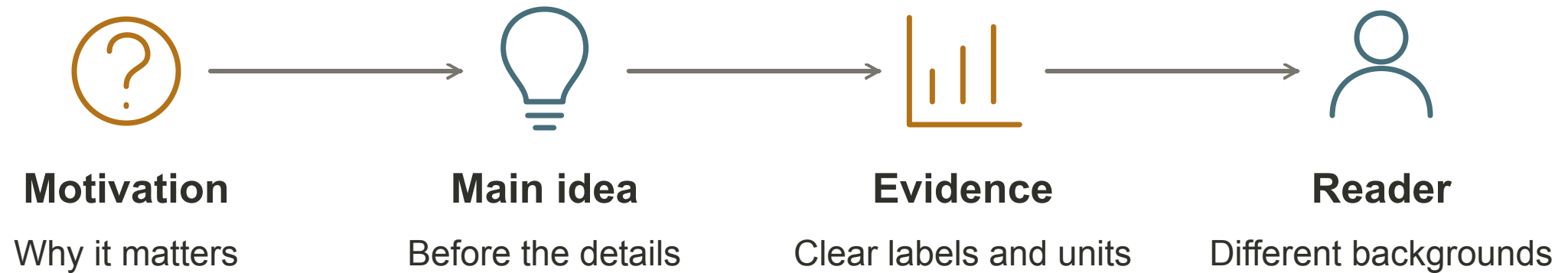
Precise claim

“In this test, our method
used 10% less energy
than the baseline.”

Hypothetical writing example, not a research result.

Explain the workload, measurement method, and variation.

Writing for your readers



Use consistent names and symbols.

Conferences and journals



Conference

Often a fixed deadline
Scheduled event
Event-based review stages



Journal

Often year-round submission
Published in a journal
May have revision rounds

Page limits, review time, and procedures vary.

Read the current venue instructions.

Choosing a publication venue

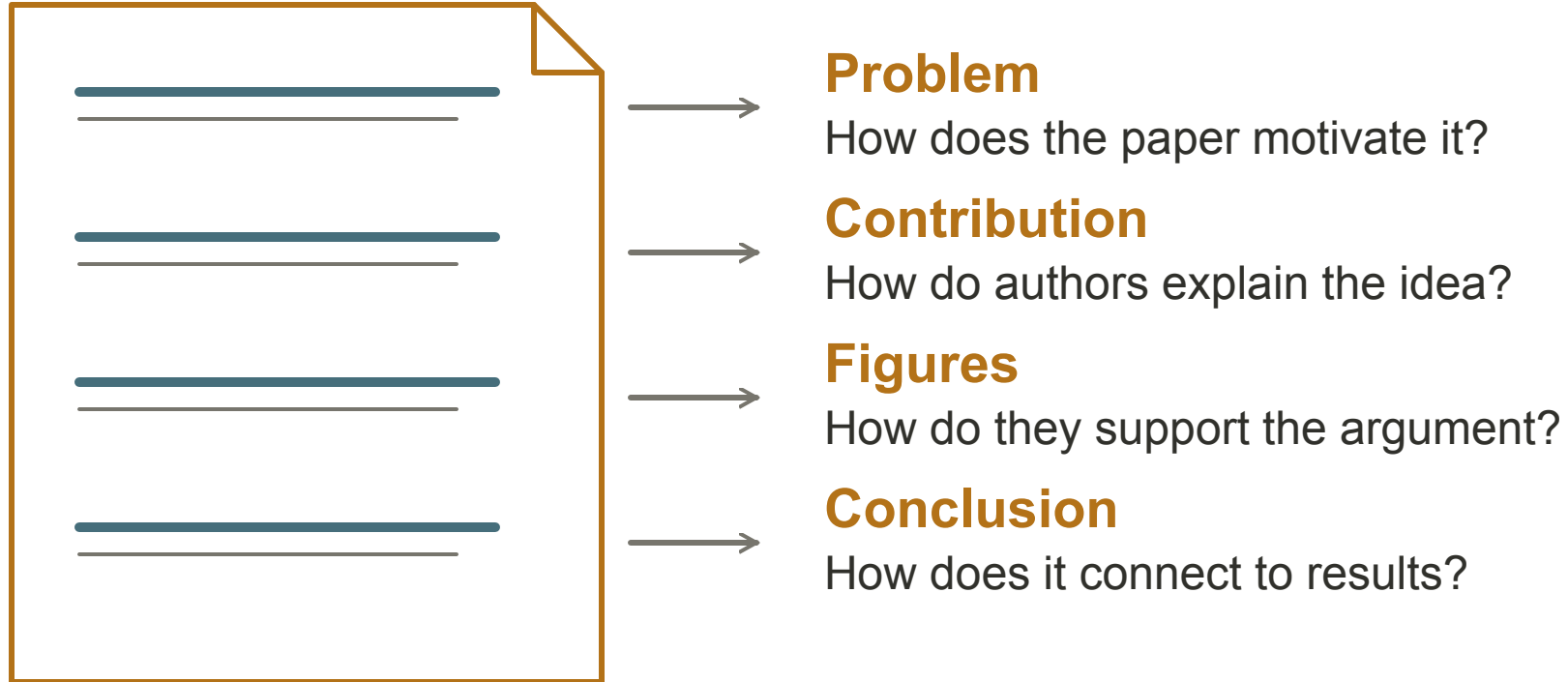
Discuss the choice with your supervisor.

- Does the venue publish work on your topic?
- Who is the intended audience?
- Are the review process and publisher clear?
- Can you meet the format and deadline?
- Are publication costs and policies acceptable?

Use rankings as one source of information.

Read recent papers and assess their relevance.

Learning from published papers



Learn from the structure. Write your own explanation.

Final/Midterm assignment



Your final draft must address concerns from all three midterm reviewers.

Individual work and course rules

The AY2026 rules:

- Individual authorship and permitted help
- Citation and reuse of research material
- Generative AI use and disclosure
- Submission format and review responsibilities

Ask the instructors if a rule is unclear.

Questions and review



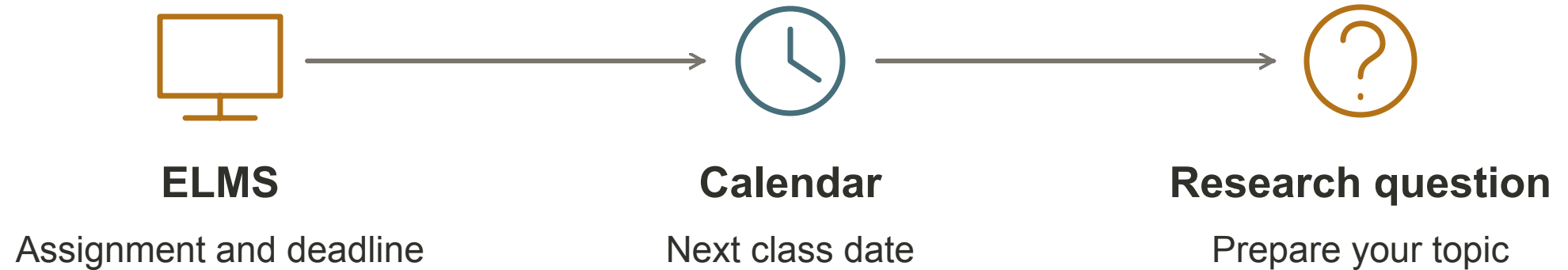
What would you like to ask about the course?

Four points to remember

- Be precise.
- Use simple language.
- Improve the paper through revision.
- Focus on quality rather than length.



Preparation for Session 02



Next: Literature survey and review

Thank you

Questions?

Research Paper Writing 1

Session 01, AY2026

Next topic: Literature survey and review